1. Code:

```cpp
#include <iostream>
#include <string>

using namespace std;

class Student {
private:
    string name;
    int rollNumber;
    float marks;

public:
    Student(string studentName, int studentRoll, float studentMarks) {
        name = studentName;
        rollNumber = studentRoll;
        marks = studentMarks;
        cout << "Constructor called: Memory allocated and details initialized for " << name << ".\n";
    }

    void displayDetails() {
        cout << "\n Student Details \n";
        cout << "Name: " << name << "\n";
        cout << "Roll Number: " << rollNumber << "\n";
        cout << "Marks: " << marks << "%\n";
    }
    ~Student() {
        cout << "Destructor called: Memory freed and object destroyed for " << name << ".\n";
    }
};

int main() {
    cout << "Starting Program Execution...\n\n";
    {
        Student student1("John Doe", 101, 89.5);
        student1.displayDetails();

    }

    cout << "Program Execution Completed.\n";

    return 0;
}
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

amol@Amols-MacBook-Pro PP-lab % g++ exp1/program1.cc
amol@Amols-MacBook-Pro PP-lab % ./a.out
Starting Program Execution...

Constructor called: Memory allocated and details initialized for John Doe.

 Student Details
Name: John Doe
Roll Number: 101
Marks: 89.5%
Destructor called: Memory freed and object destroyed for John Doe.
Program Execution Completed.
amol@Amols-MacBook-Pro PP-lab %
```

2. Code:

```
#include <iostream>

using namespace std;

class Shape {
public:
    virtual void calculateAndDisplayArea() const {
        cout << "Area of generic shape\n";
    }
    virtual ~Shape() {}
};

class Rectangle : public Shape {
private:
    float length, width;
public:
    Rectangle(float l, float w) : length(l), width(w) {}
    void calculateAndDisplayArea() const override {
        cout << "Rectangle Area: " << (length * width) << "\n";
    }
};

class Circle : public Shape {
private:
    float radius;
public:
    Circle(float r) : radius(r) {}
    void calculateAndDisplayArea() const override {
        cout << "Circle Area: " << (3.14159f * radius * radius) << "\n";
    }
};

int main() {
    float r, l, w;

    cout << "Enter length and width of the rectangle: ";
    cin >> l >> w;
    Shape* shapePointer1 = new Rectangle(l, w);

    cout << "Enter radius of the circle: ";
    cin >> r;
    Shape* shapePointer2 = new Circle(r);

    cout << "\n  Results   \n";
    shapePointer1->calculateAndDisplayArea();
    shapePointer2->calculateAndDisplayArea();

    delete shapePointer1;
    delete shapePointer2;

    return 0;
}
```

Output:

```
amol@Amols-MacBook-Pro PP-lab % g++ ./src/exp1/program2.cc
amol@Amols-MacBook-Pro PP-lab % ./a.out
Enter length and width of the rectangle: 12 32
Enter radius of the circle: 24

  Results
Rectangle Area: 384
Circle Area: 1809.56
amol@Amols-MacBook-Pro PP-lab %
```